



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



Steam Workshop abused to spread malware via Wallpaper Engine app

Malware Analysis Report

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TLP:CLEAR | Lost Boys Cyber | Steam Workshop abused to spread malware via Wallpaper Engine app - Malware Analysis Report

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Steam Workshop Abused to Spread Malware via Wallpaper Engine App

1. Executive Summary

Threat actors are abusing Steam Workshop distribution for Wallpaper Engine “application wallpapers” to deliver commodity malware, credential stealers, backdoors, crypto miners, and downloader components to Windows users. The strongest technical source is Kaspersky Securelist reporting from 16 June 2026, which identifies malicious Wallpaper Engine Workshop items observed since late 2025 and describes one infection chain that drops DarkKomet-family backdoor tooling, launches a benign-looking game wallpaper, loads a modified AggregatorHost.dll, searches for Steam session material, and exfiltrates collected data to attacker-controlled infrastructure.

This report is source-backed rather than sample-backed: Lost Boys Cyber did not acquire or detonate the Wallpaper Engine packages locally during this review. As a result, static, code, and dynamic analysis sections distinguish vendor-observed facts from defender-oriented assessment. Kaspersky’s published indicators include 13 MD5 hashes, direct IP-based C2 endpoints, cloud-download URLs, and multiple Steam Workshop item IDs that had reportedly been removed by Steam before publication. BleepingComputer corroborated the same campaign theme and highlighted operational impact: hijacked Steam accounts, system compromise through backdoors, and cryptomining activity from infected wallpapers.

The risk is not limited to gaming endpoints. Any enterprise that permits Steam, Wallpaper Engine, or unmanaged Workshop content on corporate or contractor systems should treat application wallpapers as untrusted executable content. The most urgent controls are to restrict or remove Wallpaper Engine on managed Windows hosts, block known infrastructure, hunt for direct execution from Steam Workshop content paths, monitor credential/session theft behavior, and reset Steam and browser credentials for any endpoint where a malicious wallpaper was installed.

Key Finding	Analyst Assessment	Defensive Implication
Steam Workshop acted as a malware distribution channel	High confidence from Kaspersky primary reporting and BleepingComputer secondary reporting	Trust in app-store/community distribution should not replace endpoint inspection for executable Workshop content
Wallpaper Engine application wallpapers can run Windows-compatible third-party programs	Directly documented by Kaspersky and inherent to the Wallpaper Engine feature model	Application wallpapers should be governed like downloaded executables, not media files
Observed payload set includes DarkKomet, Lumma, Vidar, RenEngine, miners, ransomware-class detections, and Python droppers	High-confidence vendor observation; individual samples were not locally acquired	Detection should cover both delivery behavior and payload families, not one hash or one malware name
Primary observed targeting centered on China and Russia	Kaspersky telemetry reported 89% of malicious downloads in China and 5.5% in Russia	Global organizations should still hunt because the delivery mechanism can be reused in any region

2. Sample Metadata

Field	Value
Family	Multi-family distribution cluster: DarkKomet, Lumma, Vidar, RenEngine, miners, Python droppers, ransomware-class payloads
Malware Type	Steam Workshop-distributed Trojanized Wallpaper Engine application wallpaper packages; stealer/backdoor/loader/miner mix
Platform	Windows endpoints running Steam and Wallpaper Engine
Primary Sample SHA256	Not published in reviewed sources; no local sample acquired

Additional Hashes	13 MD5 values published by Kaspersky; see Section 11 and iocs.csv	
File Type / Format	Wallpaper Engine Workshop content packages containing Windows executables, DLLs, scripts, and archives	
Delivery Theme	Benign-looking live/game wallpapers, including application wallpapers that run as desktop background programs	
First Seen / Reported	Activity observed since late 2025; public reporting published 2026-06-16	
Analyst Confidence	Medium-high for campaign mechanics; medium for per-sample behavior due to no local detonation	
Primary Objective	Steam account/session theft, broader credential theft, backdoor access, payload staging, and monetization through mining or additional compromise	
Published MD5	Source Context	Notes
95856f2ce428c728d9781d3296558068	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
af080780cca2acd1d082ce01e7cc346a	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
c133c3dd9f7d6934598025047df41abf	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
d1693bbff456ae8fa3360446706df6da	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
8c2cc585ad8a13a72a704c0fda0c9854	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
b9fa763a53da3eea742d0f3c845a8c09	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
ded08ae5df7f1b12e5fdb767dbbed0b1	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
20965254e29104986e11939decd39549	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
18dedc0009f0927cba6425c84cce9883	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
0f4f01c6d495abb37403072dd017ce8d	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
5620f01284329f561b1839a36be55355	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
fe1f6485013cd5e6d5cf718049b0b8d6	Kaspersky IOC list	Related malicious object; exact file role not mapped in public source
74414ed4b63aadec039b603c32762b80	Kaspersky IOC list	Related malicious object; exact file role not

		mapped in public source
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3. Source Review & Confidence

Source	Contribution	Confidence
Kaspersky Securelist, “Dozens of malicious wallpapers found on Steam Workshop: gamers’ accounts at risk,” 2026-06-16	Primary technical narrative, infection-chain description, targeting telemetry, malware families, C2 URLs, MD5s, and Steam Workshop item IDs	High for reported vendor observations
BleepingComputer, “Steam Workshop abused to spread malware via Wallpaper Engine app,” 2026-06-16	Secondary corroboration, public impact framing, and defender-facing summary	Medium-high
Lost Boys Cyber local analysis	Source review, detection engineering, ATT&CK mapping, hunt logic, and defensive interpretation	Medium; no local sample acquisition

Directly observed evidence in this report means evidence observed by named external researchers and preserved in their public reporting. High-confidence assessment includes conclusions that follow from those observations, such as treating application wallpapers as executable content and scoping Steam session theft. Lower-confidence inference includes exact per-hash capability mapping, because the public source provided MD5 values and family-level verdicts but not a full sample-by-sample reverse-engineering matrix.

The public reporting states that Steam removed identified malicious wallpapers and links before publication. That reduces immediate exposure to the exact listed Workshop items, but does not remove the delivery-channel risk: any similar Workshop package can reappear under a new item ID, archive name, or payload hash.

4. Initial Triage and Delivery Assessment

Wallpaper Engine supports multiple wallpaper classes: videos, scenes, web pages, and application wallpapers. The threat campaign focuses on application wallpapers because they are effectively Windows-compatible programs rendered as desktop backgrounds. Kaspersky observed attackers publishing malicious application wallpapers to Steam Workshop, where users could download and apply them like community content.

Two delivery patterns were reported:

Variant / Artefact	Delivery Role	Notes
Archive with executable wallpaper plus malicious files	Direct payload delivery	Archive may contain EXE, DLL, or malicious scripts staged beside a functional wallpaper
Password-protected archive with embedded or obvious password	Evasion / user-assisted unpacking	Password appeared in archive names or JSON configuration, or scripts handled extraction automatically
Benign-looking game wallpaper	Social engineering cover	Sample described by Kaspersky launched a functioning game, NTRaholic, while infection ran in the background
._cache_GAME1.exe	Game launcher and installer component	Launched the game and installed a modified AggregatorHost.dll payload
Synaptics.exe	Dropped backdoor	Reported as part of the DarkKomet malware family

AggregatorHost.dll	Credential/session theft component	Modified library searched for Steam application data and hijacked a live Steam session
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Operationally, the lure is effective because a user expects Workshop content to create new files and run inside Wallpaper Engine. That expectation lowers the chance that a game wallpaper, helper process, or unusual content directory will be reported quickly by the user.

5.1 File and Build Characteristics

No local binary was available for static tooling, so the static analysis below is derived from vendor-reported artifact names, delivery formats, and IOCs. The most important static triage conclusion is that the package should not be treated as a media-only artifact: the campaign relies on executable content and native Windows components stored inside Workshop-distributed wallpaper packages.

Static Signal	Evidence	Analyst Value
Executable application wallpaper	Wallpaper Engine application wallpaper feature runs third-party Windows-compatible software	Controls must classify selected Wallpaper Engine packages as executable code
Dropped Synaptics.exe	Kaspersky reported a DarkKomet-family backdoor dropped under this name	Filename masquerades as a legitimate input-device vendor component and should be triaged by path/signature/hash
Modified AggregatorHost.dll	Reported custom system-library impersonation carrying credential/session theft logic	DLL name may blend with Windows system components; investigate non-system-path copies and unsigned variants
._cache_GAME1.exe	Reported module that launches NTRaholic and installs the modified DLL	Suspicious hidden/cache-prefixed executable in Workshop content path
Password-protected archives	Kaspersky observed passwords in archive names or JSON config	Password-protected packaging can suppress static inspection by mail/web gateways and some endpoint tooling
Direct IP C2	120.48.156.17 and 202.144.192.29 appear in published infrastructure	Direct-IP HTTP is high-signal on managed endpoints, especially from game/wallpaper process lineage

5.2 Strings, Imports, and Embedded Artefacts

Published reporting did not provide full strings, imports, PE metadata, compiler timestamps, resources, or section entropy. Defenders acquiring local copies should prioritize extraction of:

- Archive member lists, including EXE/DLL/script names and passwords embedded in names or JSON.
- Wallpaper Engine project metadata and JSON configuration that references executable entry points.
- PE signing state and original filenames for Synaptics.exe, ._cache_GAME1.exe, AggregatorHost.dll, and other adjacent binaries.
- Hard-coded URLs, IP addresses, and paths matching the C2 indicators in Section 10.
- Steam credential/session path references, Steam client process discovery logic, and browser credential store access.

5.3 Static Configuration Clues

The public C2 list includes direct HTTP endpoints and cloud-storage staging URLs:

Configuration Clue	Published Value	Assessment
Steam-session exfiltration endpoint	hxxp://120.48.156[.]17/ey.php	Reported destination for data collected by

		AggregatorHost.dll
Additional server and audit endpoint	hxxp://202.144.192[.]29 and /audit.php	Likely staging, telemetry, or secondary command endpoint
Theme archive URL	hxxp://202.144.192[.]29/download2/Themes2.zip and hxxp://brightly[.]to/download2/Themes2.zip	Payload/theme staging pattern worth blocking and hunting
Dropbox-hosted RAR	hxxps://www.dropbox[.]com/s/zhp1b06imehwylq/Synaptics.rar?dl=1	Cloud-hosted payload retrieval; defenders should preserve URL and downloaded object hash if seen
Google Docs download URL	hxxps://docs.google[.]com/uc?id=0BxsMXGfPIZfSVzUyaHFYVkQxeFk&export=download	Cloud-hosted payload retrieval; detect from suspicious parent process rather than domain alone

6.1 Reverse-Engineering Method

Lost Boys Cyber did not receive a sample package for Ghidra, radare2, capa, FLOSS, or sandbox validation. This section documents the code-level behavior reported by Kaspersky and translates it into review targets for defenders who recover samples internally.

Method	Status	Result
Public-source code behavior review	Completed	Kaspersky narrative identifies dropper, game launcher, modified DLL, Steam discovery, session hijack, and exfiltration
Local decompilation	Not performed	No sample available during this task
Local config extraction	Not performed	Public URLs and MD5s were preserved in iocs.csv
Detection logic derivation	Completed	Sigma, YARA-style heuristics, KQL, and SPL included for defender adaptation

6.2 Functions and Logic of Interest

Function / Routine	Evidence	Why It Matters
Wallpaper launch handler	Malicious payload triggers when the user selects/applies the wallpaper	Execution may be user-initiated through Wallpaper Engine rather than a traditional installer
Drop Synaptics.exe	Kaspersky reported the backdoor drop into the victim system	Establishes secondary malware execution independent of the visible wallpaper
Launch ._cache_GAME1.exe	Reported executable starts the NTRaholic game	Provides cover traffic and user-facing legitimacy
Install modified AggregatorHost.dll	Reported by Kaspersky as part of the same module's double duty	Enables library impersonation and credential/session theft behavior
Locate Steam application/session	Kaspersky screenshots and narrative describe searching for Steam app and hijacking the session	Supports account takeover and malicious re-use of victim Steam identity

HTTP exfiltration to ey.php	Public report states collected data is sent to attacker server	High-value network detection and containment trigger
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6.3 Anti-Analysis, Evasion, and Obfuscation

The public reporting identifies several evasion-adjacent choices even without a full binary analysis:

- Password-protected archives can prevent automated content inspection and require password recovery from filenames or JSON configuration.
- Legitimate-looking wallpaper/game behavior masks background installation.
- DLL naming as AggregatorHost.dll and executable naming as Synaptics.exe can blend with familiar Windows or vendor terminology.
- Steam Workshop hosting provides reputational cover and a trusted user acquisition path.
- Multiple unrelated malware families indicate either many operators or copycat reuse, which reduces the value of one-family signatures.

7.1 Execution Overview

Vendor-observed execution begins when a user applies an infected application wallpaper. The visible component launches normally and displays the expected wallpaper/game. In the background, the package drops and runs malicious components that install a backdoor, search for Steam session data, and communicate with attacker infrastructure.

7.2 Filesystem, Registry, Service, and Task Changes

Public reporting did not describe registry persistence, services, scheduled tasks, WMI, or autoruns for every variant. Because the payload set includes DarkKomet, infostealers, loaders, miners, and ransomware-class detections, incident responders should assume that persistent changes may vary by downloaded payload.

Change Type	Artefact	Evidence	Confidence
File drop	Synaptics.exe	Kaspersky report describes DarkKomet-family backdoor drop	High for described sample
File execution	._cache_GAME1.exe	Kaspersky report describes game-launching module	High for described sample
DLL placement/loading	AggregatorHost.dll	Kaspersky report describes custom library installation	High for described sample
Archive extraction	Password-protected archive content	Kaspersky report describes archive and password handling	Medium-high across variants
Persistence	Unknown / payload-dependent	Not specified in public report	Low until local sample or telemetry confirms

7.3 Network and Command-and-Control Behaviour

The most explicit network behavior is HTTP exfiltration to 120.48.156.17/ey.php after the modified DLL collects Steam session data. Other published URLs suggest payload staging and infrastructure reuse across variants.

Network Behaviour	Evidence	Defensive Notes
Direct-IP HTTP request to 120.48.156.17/ey.php	Kaspersky narrative and IOC list	Block, alert, and scope any endpoint with this connection
HTTP access to 202.144.192.29/audit.php and /download2/Themes2.zip	Kaspersky IOC list	Investigate as possible staging or campaign telemetry
brightly.to download path	Kaspersky IOC list	Domain may serve theme/payload archives;

		inspect downloaded ZIP/RAR content
Dropbox and Google Docs downloads	Kaspersky IOC list	Parent process and command line are more reliable than domain-only blocks

7.4 Observed Capability Summary

Capability	Supporting Dynamic Evidence
Steam session hijacking	Modified AggregatorHost.dll searches for Steam app/session material and sends collected data to attacker server
Backdoor installation	Synaptics.exe reported as DarkKomet-family backdoor dropped during wallpaper launch
Payload staging	Public C2 and cloud-download URLs indicate secondary content retrieval
Credential theft	Kaspersky detections include Trojan-PSW verdicts; Lumma and Vidar families are named
Resource monetization	Reporting notes crypto miners distributed through the same technique
Additional compromise enablement	Backdoor, loader, and ransomware-class verdicts support broader intrusion risk

8. Process Tree and Execution Chain

A representative process chain for the Kaspersky-described sample is below. Process names and paths should be adjusted during local triage because exact install paths vary by Steam library location and Workshop item ID.

Step	Parent Process	Child / Action	Why It Matters
1	steam.exe / Wallpaper Engine process	User downloads and applies a Steam Workshop application wallpaper	User action launches executable content from a trusted community-content workflow
2	wallpaper executable	Benign-looking game wallpaper starts	Visual cover reduces suspicion and supports longer dwell time
3	wallpaper executable	Drops Synaptics.exe	Establishes DarkKomet-family backdoor component
4	wallpaper executable	Starts ._cache_GAME1.exe	Launches NTRaholic game and performs installer role
5	._cache_GAME1.exe	Installs or loads modified AggregatorHost.dll	Enables credential/session theft under a system-like DLL name
6	AggregatorHost.dll-hosting process	Locates Steam application/session material	Supports Steam account hijacking
7	AggregatorHost.dll-hosting process	HTTP POST/GET to 120.48.156.17/ey.php	Exfiltrates collected data to attacker-controlled server
8	Attacker infrastructure	Uses victim Steam account to publish or amplify malicious	Turns account takeover into

		wallpapers	propagation capability
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Potential Windows paths to inspect include:

C:\Program Files (x86)\Steam\steamapps\workshop\content\431960\<workshop_item_id>\
C:\Program Files (x86)\Steam\steamapps\common\wallpaper_engine\
<alternate SteamLibrary>\steamapps\workshop\content\431960\<workshop_item_id>\

9. Persistence and System Modification

The publicly described sample clearly modifies the host by dropping and installing executable/DLL components, but it does not provide enough detail to assert a specific persistence key or service. DarkKomet and commodity loaders commonly support persistence, but that should be validated per recovered payload.

Modification Type	Artefact	Evidence	Confidence
Dropped executable	Synaptics.exe	Kaspersky report	High
Modified library	AggregatorHost.dll	Kaspersky report	High
Game launcher / cache executable	._cache_GAME1.exe	Kaspersky report	High
Steam account/session compromise	Active Steam session data	Kaspersky report	High
Startup persistence	Registry Run keys, scheduled tasks, services, startup folders	Not published; hunt recommended	Medium as a payload-dependent possibility
Account abuse	Victim Steam account used to upload more malicious wallpapers	Kaspersky assessment	Medium-high

Responders should collect autoruns, scheduled tasks, services, loaded modules, Amcache/Shimcache, Prefetch, SRUM, Windows Defender history, browser credential access telemetry, and Steam login/session artifacts before remediation.

10. Network and Infrastructure Analysis

IOC / Infrastructure	Type	Role	Stability / Confidence
120.48.156.17	IP	C2 / exfiltration server	High confidence; may be short-lived
hxxp://120.48.156[.]17/ey.php?ka=user1&id	URL pattern	Steam/session data exfiltration	High confidence from Kaspersky IOC list
202.144.192.29	IP	Server for audit/download paths	High confidence; may rotate
hxxp://202.144.192[.]29/audit.php	URL	Possible telemetry/audit endpoint	Medium-high
hxxp://202.144.192[.]29/download2/Themes2.zip	URL	Staging/download archive	Medium-high
brightly[.]to	Domain	Staging/download infrastructure	Medium-high
hxxp://brightly[.]to/download2/Themes2.zip	URL	Staging/download archive	Medium-high
hxxps://www.dropbox[.]com/	URL	Cloud-hosted RAR payload	Medium-high; cloud content

s/zhp1b06imehwylq/ Synaptics.rar?dl=1			may disappear
hxxps://docs.google[.]com/uc? id=0BxsMXGfPIZfSVzUyaHF YVkJxeFk&export=download	URL	Cloud-hosted payload retrieval	Medium-high; domain not malicious by itself
Steam Workshop item IDs listed in Section 11	URL	Malicious wallpaper distribution pages	High confidence historically; reportedly removed before publication

Infrastructure includes both direct attacker-controlled endpoints and legitimate cloud-hosted download mechanisms. Block direct IPs and known malicious URLs where possible, but avoid broad blocks on Dropbox or Google Docs without process lineage and URL specificity because those services have legitimate use.

11. Indicators of Compromise

Machine-readable indicators are preserved in iocs.csv. The table below summarizes the indicator set for triage and detection engineering.

IOC Type	Example / Count	Source	Operational Notes
MD5	13 hashes	Kaspersky	Use for retrospective scanning where MD5 telemetry exists; request SHA256 from local samples when possible
IP	2 IP addresses	Kaspersky	Direct-IP HTTP from Wallpaper Engine, Steam, or unusual child processes is high signal
URL	7 URL patterns	Kaspersky	Includes C2, staging ZIP/RAR, Dropbox, and Google Docs download URLs
Domain	brightly.to plus cloud-service URLs	Kaspersky	Hunt exact URLs and parent process lineage rather than broad cloud-domain blocks
Steam Workshop Item	14 item URLs	Kaspersky	Reportedly removed, but useful for user interview, Steam cache review, and browser history checks
File Names	Synaptics.exe, _cache_GAME1.exe, AggregatorHost.dll	Kaspersky	Names are not unique enough alone; combine with path, signer, hash, and process lineage

High-value published indicators:

120.48.156.17 202.144.192.29 hxxp://120.48.156[.]17/ey.php?ka=user1&id hxxp://202.144.192[.]29/audit.php hxxp://202.144.192[.]29/download2/Themes2.zip hxxp://brightly[.]to/download2/Themes2.zip hxxps://www.dropbox[.]com/s/zhp1b06imehwylq/Synaptics.rar?dl=1 hxxps://docs.google[.]com/uc?id=0BxsMXGfPIZfSVzUyaHFYVkJxeFk&export=download
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12. MITRE ATT&CK Mapping

Technique	Name	Evidence
T1195.002	Supply Chain Compromise: Compromise Software Supply Chain	Steam Workshop community-content distribution used to deliver malicious Wallpaper Engine packages
T1204.002	User Execution: Malicious File	User applies an application wallpaper that executes malicious content
T1105	Ingress Tool Transfer	Published cloud and HTTP download URLs indicate staged payload retrieval
T1059	Command and Scripting Interpreter	Kaspersky reported malicious scripts among payload delivery methods; specific interpreter varies by variant
T1574.001	Hijack Execution Flow: DLL Search Order Hijacking	Modified AggregatorHost.dll installed/loaded to perform credential theft behavior; exact load mechanism requires local validation
T1555	Credentials from Password Stores	Lumma/Vidar and Trojan-PSW verdicts indicate credential-theft objectives
T1539	Steal Web Session Cookie	Steam active-session hijack maps most closely to theft of session material
T1071.001	Application Layer Protocol: Web Protocols	HTTP C2/exfiltration to ey.php and related infrastructure
T1041	Exfiltration Over C2 Channel	AggregatorHost.dll sends collected data to attacker-controlled server
T1090 / T1102	Proxy or Web Service Use	Dropbox and Google Docs URLs may be used for payload staging through legitimate services
T1496	Resource Hijacking	Reporting described cryptomining payloads delivered through malicious wallpapers
T1027	Obfuscated Files or Information	Password-protected archives and disguised executable names complicate inspection

13.1 Detection Logic Summary

Signal	Telemetry Source	Analytic Goal	Tuning Notes
Executable launch from Wallpaper Engine Workshop content	EDR process events	Identify application wallpapers launching EXE/DLL/script payloads	Baseline legitimate Wallpaper Engine app-wallpaper use; prioritize unusual filenames and network connections
Synaptics.exe outside vendor paths	EDR process/file events	Detect masqueraded DarkKomet dropper name	Exclude signed Synaptics drivers in legitimate system/vendor directories
AggregatorHost.dll outside	Module/file events	Detect malicious DLL	Tune for OS version and

System32/SysWOW64 or loaded by non-standard parent		impersonation	legitimate AggregatorHost artifacts
Direct HTTP to 120.48.156.17 or 202.144.192.29	DNS/proxy/firewall/EDR network events	Detect C2/staging activity	Treat exact direct-IP contact as high severity
Cloud downloads of Synaptics.rar or listed Google Docs ID	Proxy/EDR network events	Detect secondary payload retrieval	Do not block all Dropbox/Google Docs; alert on exact URL and suspicious parent process
Steam Workshop item IDs in browser/cache/Steam logs	Browser history, Steam cache, filesystem	Scope affected users	Item IDs may be removed; preserve local cache evidence

13.2 Detection Content

```

title: Suspicious Wallpaper Engine Workshop Executable Launch
id: b7e68f96-3aa5-4d2b-a421-steam-wallpaper-engine-workshop-exec
status: experimental
description: Detects executable or script launch from Steam Workshop content paths used by Wallpaper Engine application wallpapers.
author: Lost Boys Cyber
date: 2026-06-17
references:
  - https://securelist.com/dozens-of-malicious-wallpapers-found-on-steam-workshop/120186/
  - https://www.bleepingcomputer.com/news/security/steam-workshop-abused-to-spread-malware-via-wallpaper-engine-app/
logsource:
  product: windows
  category: process_creation
detection:
  selection_path:
    Image|contains:
      - '\\steamapps\\workshop\\content\\431960\\'
      - '\\SteamLibrary\\steamapps\\workshop\\content\\431960\\'
  selection_ext:
    Image|endswith:
      - '.exe'
      - '.cmd'
      - '.bat'
      - '.ps1'
      - '.vbs'
      - '.js'
      - '.wsf'
  suspicious_names:
    Image|endswith:
      - '\\Synaptics.exe'
      - '\\._cache_GAME1.exe'
      - '\\AggregatorHost.exe'
  condition: selection_path and (selection_ext or suspicious_names)
falsepositives:
  - Legitimate Wallpaper Engine application wallpapers can run executables. Baseline approved Workshop items and publisher-controlled endpoints.
level: high

```

```

title: Wallpaper Engine Related Connection To Published Malware Infrastructure
id: 394be8b0-baae-46a7-9cfd-steam-wallpaper-engine-c2
status: experimental
description: Detects network connections to published Steam Workshop / Wallpaper Engine malware infrastructure from Windows endpoints.
author: Lost Boys Cyber
date: 2026-06-17
logsource:
  product: windows

```

```

category: network_connection
detection:
  selection_ip:
    DestinationIp:
      - '120.48.156.17'
      - '202.144.192.29'
  selection_proc:
    Image|contains:
      - '\\steamapps\\workshop\\content\\431960\\'
      - '\\wallpaper_engine\\'
      - '\\steam.exe'
  condition: selection_ip and selection_proc
falsepositives:
  - None expected for exact listed IPs; investigate any hit.
level: critical

rule LBC_Steam_Workshop_WallpaperEngine_Suspicious_Package_Heuristic {
  meta:
    description = "Heuristic for suspicious Wallpaper Engine application wallpaper packages referencing published Steam Workshop malware artifacts"
    author = "Lost Boys Cyber"
    date = "2026-06-17"
    reference = "https://securelist.com/dozens-of-malicious-wallpapers-found-on-steam-workshop/120186/"
    confidence = "medium"
  strings:
    $we_path = "steamapps\\workshop\\content\\431960" nocase
    $syn = "Synaptics.exe" nocase
    $cache = "._cache_GAME1.exe" nocase
    $agg = "AggregatorHost.dll" nocase
    $ip1 = "120.48.156.17"
    $ip2 = "202.144.192.29"
    $ey = "/ey.php" nocase
    $themes = "/download2/Themes2.zip" nocase
  condition:
    uint16(0) == 0x5a4d and (2 of ($syn,$cache,$agg,$ip1,$ip2,$ey,$themes) or ($we_path and 1 of ($syn,$cache,$agg,$ip1,$ip2,$ey,$themes)))
}

index=edr sourcetype=process
(Image="*\\steamapps\\workshop\\content\\431960\\*" OR CommandLine="*\\steamapps\\workshop\\content\\431960\\*")
(Image="*.exe" OR Image="*.cmd" OR Image="*.bat" OR Image="*.ps1" OR Image="*.vbs" OR Image="*.js" OR Image="*\\Synaptics.exe" OR Image="*\\._cache_GAME1.exe")
| stats min(_time) as firstSeen max(_time) as lastSeen values(ParentImage) as parents
values(CommandLine) as commandLines values(User) as users by host Image
| convert ctime(firstSeen) ctime(lastSeen)

```

13.3 Detection Validation and Blind Spots

Validate these detections against known-good Wallpaper Engine usage before broad deployment. Some organizations may legitimately permit application wallpapers that run helper executables. Prioritize detections when process execution from Workshop paths is followed by direct-IP network connections, DLL placement under non-system paths, credential-store access, or child processes with hidden/cache-like names.

Blind spots include renamed payloads, new Workshop item IDs, HTTPS-only infrastructure, payloads that execute from extracted temp directories rather than the Workshop path, and variants that rely on cloud services without direct-IP C2. Complement path and IOC rules with behavioral analytics for credential theft and unsigned executable execution from user-writable content directories.

14.1 Hunt Hypotheses

- A user applied a malicious Wallpaper Engine application wallpaper that launched an executable or script from a Steam Workshop content directory.
- A malicious wallpaper package dropped Synaptics.exe, ._cache_GAME1.exe, or AggregatorHost.dll outside expected trusted locations.
- A compromised endpoint contacted published C2 or staging infrastructure after Steam/Wallpaper Engine activity.

- A victim Steam account may have been hijacked and used to upload or interact with additional malicious Workshop content.

14.2 Hunt Queries

```
// Microsoft Defender XDR: executable/script launch from Wallpaper Engine Workshop content
DeviceProcessEvents
| where Timestamp > ago(60d)
| where FolderPath has_any ("\\steamapps\\workshop\\content\\431960\\", "\\SteamLibrary\\steamapps\\workshop\\content\\431960\\")
| where FileName endswith ".exe" or FileName endswith ".cmd" or FileName endswith ".bat" or FileName endswith ".ps1" or FileName endswith ".vbs" or FileName endswith ".js" or FileName in~ ("Synaptics.exe", ".cache_GAME1.exe")
| project Timestamp, DeviceName, InitiatingProcessFileName, InitiatingProcessCommandLine, FileName, FolderPath, ProcessCommandLine, AccountName, SHA256, MD5
| order by Timestamp desc

// Microsoft Defender XDR: published C2/staging infrastructure from Steam or Wallpaper Engine lineage
let SteamWallpaperTerms = dynamic(["steam.exe", "wallpaper32.exe", "wallpaper64.exe", "wallpaper_engine", "steamapps\\workshop\\content\\431960"]);
DeviceNetworkEvents
| where Timestamp > ago(60d)
| where RemoteIP in ("120.48.156.17", "202.144.192.29") or RemoteUrl has_any ("120.48.156.17/ey.php", "202.144.192.29/audit.php", "202.144.192.29/download2/Themes2.zip", "brightly.to/download2/Themes2.zip", "zhp1b06imehwylq/Synaptics.rar", "0BxsMXGfPIZfSVzUyaHFYVvkQxeFk")
| where InitiatingProcessFileName has_any (SteamWallpaperTerms) or InitiatingProcessCommandLine has_any (SteamWallpaperTerms) or InitiatingProcessFolderPath has_any (SteamWallpaperTerms)
| project Timestamp, DeviceName, InitiatingProcessFileName, InitiatingProcessFolderPath, InitiatingProcessCommandLine, RemoteIP, RemoteUrl, RemotePort, Protocol, AccountName
| order by Timestamp desc

// Microsoft Defender XDR: suspicious file writes for reported artifact names
DeviceFileEvents
| where Timestamp > ago(60d)
| where FileName in~ ("Synaptics.exe", ".cache_GAME1.exe", "AggregatorHost.dll")
| where FolderPath !startswith "C:\\Windows\\System32" and FolderPath !startswith "C:\\Windows\\SysWOW64"
| project Timestamp, DeviceName, ActionType, FileName, FolderPath, InitiatingProcessFileName, InitiatingProcessCommandLine, SHA256, MD5, AccountName
| order by Timestamp desc

// Microsoft Defender XDR: known MD5s from public IOC list
let md5s = dynamic(["95856f2ce428c728d9781d3296558068", "af080780cca2acd1d082ce01e7cc346a", "c133c3dd9f7d6934598025047df41abf", "d1693bbff456ae8fa3360446706df6da", "8c2cc585ad8a13a72a704c0fda0c9854", "b9fa763a53da3eea742d0f3c845a8c09", "ded08ae5df7f1b12e5fdb767dbbed0b1", "20965254e29104986e11939decd39549", "18dedc0009f0927cba6425c84cce9883", "0f4f01c6d495abb37403072dd017ce8d", "5620f01284329f561b1839a36be55355", "fe1f6485013cd5e6d5cf718049b0b8d6", "74414ed4b63aadec039b603c32762b80"]);
union DeviceFileEvents, DeviceProcessEvents
| where Timestamp > ago(90d)
| where MD5 in~ (md5s)
| project Timestamp, DeviceName, ActionType, FileName, FolderPath, ProcessCommandLine, InitiatingProcessFileName, InitiatingProcessCommandLine, MD5, SHA256, AccountName
| order by Timestamp desc
```

14.3 Triage Guidance for Hunt Hits

A meaningful hit is an executable or script launched from Wallpaper Engine Workshop content, especially if followed by contact to a listed IP/URL, creation of Synaptics.exe or AggregatorHost.dll, credential-store access, or anomalous Steam account activity. Benign lookalikes include legitimate application wallpapers that run small games, clocks, or monitoring widgets; require additional evidence before declaring compromise.

Escalate to incident response when any of the following are present:

- Exact IOC hash or exact listed infrastructure contact.

- Direct-IP HTTP from a Wallpaper Engine, Steam, or Workshop-content process.
- Synaptics.exe or AggregatorHost.dll created outside trusted vendor/system locations.
- Steam account login, inventory, Workshop publishing, or credential-change activity not initiated by the user.
- Miner behavior, ransomware notes, unexpected encryption activity, or backdoor beaconing after wallpaper execution.

15. Triage and Incident Response

Priority	Action	Rationale
Critical	Isolate endpoints with exact IOC hash or listed C2 contact	Exact indicator hits imply likely compromise, not just exposure
Critical	Reset Steam credentials and revoke active sessions/tokens for affected users	The described payload targets active Steam sessions and account takeover
High	Preserve Steam, Wallpaper Engine, browser, EDR, proxy, and filesystem artifacts before cleanup	Workshop cache and process lineage may be overwritten or removed quickly
High	Scan Steam Workshop content directories and alternate SteamLibrary paths	Users often store games/content outside the default Steam path
High	Review enterprise credential exposure from same host	Commodity stealers may collect browser, password manager, and session material beyond Steam
Medium	Remove or restrict Wallpaper Engine and Steam on managed endpoints where not business-required	Reduces recurrence and closes application-wallpaper execution path
Medium	Notify users that application wallpapers are executable content	Prevents users from treating Workshop content as passive media

Collection priorities:

- Full path listings and hashes under steamapps\workshop\content\431960.
- Prefetch, Amcache, Shimcache, SRUM, Windows Defender history, and EDR process/network timelines.
- Steam login history, account changes, Workshop upload history, and browser history referencing listed item IDs.
- Copies of suspicious archives, JSON configs, passwords, EXE/DLL files, and downloaded cloud payloads.
- Network telemetry for 120.48.156.17, 202.144.192.29, brightly.to, and exact Dropbox/Google Docs URL paths.

16. Validation and Emulation

This review was validated through public-source analysis, IOC normalization, detection engineering, and query construction. It was not validated through local sandbox execution or reverse engineering because no sample package was supplied.

Safe emulation options for defenders:

Emulation Goal	Safe Method	Expected Detection
Path-based Workshop executable launch	Create a benign signed test executable under a lab path matching steamapps\workshop\content\431960 and execute it under a controlled test user	Sigma/KQL path-launch analytics should trigger
Direct-IP IOC control validation	Use a simulator or firewall test event with RemoteIP 120.48.156.17 from a fake Wallpaper Engine process name; do not	Network IOC analytics should trigger without live beaconing

	connect to live attacker infrastructure	
File-name masquerade detection	Create benign test files named Synaptics.exe and AggregatorHost.dll in a temporary non-system path	File-event hunts should trigger and exclude legitimate system/vendor paths
Hash IOC validation	Load the published MD5 values into EDR custom indicators as detect-only or audit entries	IOC matching should alert on any real file telemetry with matching MD5

Do not download live payloads from the published infrastructure for validation. If a local sample is recovered from an affected host, detonate only in an isolated malware-analysis lab with no personal Steam credentials or production browser profiles present.

17. Recommendations

Priority	Owner	Action	Rationale
Critical	Endpoint / SOC	Block and alert on listed direct IPs and exact malicious URL paths	Published C2/staging infrastructure is high-confidence and low business value
Critical	IAM / User Support	Reset Steam credentials and revoke sessions for confirmed affected users	Steam session hijack is a core observed objective
High	Endpoint Engineering	Restrict Wallpaper Engine application wallpapers or remove Wallpaper Engine from managed systems	The abused feature executes third-party code from community content
High	SOC Detection Engineering	Deploy path, filename, network, and hash detections from Sections 13-14	IOC-only coverage will miss new Workshop item IDs; behavior rules improve resilience
High	Threat Hunting	Run 60-90 day hunts across EDR, proxy, DNS, browser history, and Steam cache	Campaign has been observed since late 2025 and may predate public reporting
Medium	Vulnerability / Asset Management	Identify endpoints with Steam and Wallpaper Engine installed	Exposure inventory is required before control enforcement
Medium	User Awareness	Warn users that application wallpapers can execute software and should not be installed from unknown publishers	Reduces user-assisted execution of future packages
Medium	Incident Response	Add Steam Workshop content directories to forensic collection playbooks	Preserves evidence that users or platform moderation may remove later

18. References and Refresh Notes

- [1] Kaspersky Securelist, “Dozens of malicious wallpapers found on Steam Workshop: gamers’ accounts at risk,” 2026-06-16. <https://securelist.com/dozens-of-malicious-wallpapers-found-on-steam-workshop/120186/>

- [2] BleepingComputer, “Steam Workshop abused to spread malware via Wallpaper Engine app,” 2026-06-16.
<https://www.bleepingcomputer.com/news/security/steam-workshop-abused-to-spread-malware-via-wallpaper-engine-app/>
- [3] Steam Community discussion page referenced during source discovery, “Steam Workshop Infrastructure Abused to Deliver Malware via Wallpaper Engine,” 2026.
<https://steamcommunity.com/app/431960/discussions/2/562533727113348472/>

Refresh guidance: re-check Steam Workshop item availability, infrastructure status, and hash reputation before client release or operational blocking. The direct IPs, cloud URLs, and Workshop item IDs may be removed or repurposed quickly. If a sample is later provided, refresh Sections 5-8 with local hashes, PE metadata, decompiler findings, sandbox telemetry, and exact process trees.